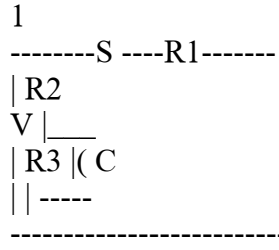
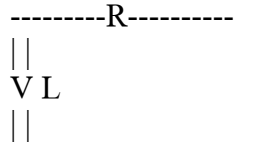
2) Some star to delta conversion networks for finding the R
3) What shud we do to reduce latch up -----
3) How to reduce short channel effects – substrate is to heavily doped
4) Some mental ability q's
5) Convert a mux to an OR gate
6) Design a 2X1 mux using half adders
7) Some clk skew q
8) Some simple ckt which has 2 voltage sources in series to it and a current source u had to find the I through resistor which is a easy one to solve
9) Two latches constructed using muxes are cascaded such that it acts like a master slave flipflop and u shud mention wether it is +ve edge triggered or –ve edge triggered..
10) Some stuck at fault in a ckt and u shud mention the test vector for it.
11) Some k map simplification….
12) Given a boolean eq. and u shud design the ckt using min no. of nmos and pmos for that go for pseudo nmos technique.
13) Given the below ckt and u shud tell wether the clk period is enough or not and what problems that the ckt will faces (I m not able to remember the correct q and diagram)
1ohm 1ohm 1ohm
1ohm 1ohm 1ohm
1ohm
6V
V_1
3V
Find V_1?
Interview questions:
Some basic inverter q's
Latch up q's
Timing violation q's
Freescale:
1. How to design AND Gate using one pMOS and one nMOS.
2. Design a flip flop using MUX.
3.Design a divide by 3 synchronous circuit.
4. Positive edge detector circuit.
5. A simple combinational circuit was asked to be simplified.
6. Design a two bit comparator with and without using MUX.
7. A transistor circuit is given.find out the output voltage given Vbe and Vce. This is a simple one.
8. Design a square wave generator which takes only one positive edge trigger.
9. A question on maximum frequency of operation of a circuit. the setup time, hold time of the flip flops are given.
f/f1
t_{setup}=3.5ns
t_{hold}=2ns
$t_{c\text{-}q}$=3ns
f/f1
t_{setup}=3.5ns
t_{hold}=2ns
$t_{c\text{-}q}$=3ns
Buffer
t_{buffer}=3.3ns
D

Clk
T_{clk}=5ns
T_{comb}=3ns
10. What is the purpose of the impedence matching between the load and source?
ans: To avoid the reflection of the power.
ITTIAM:
Written Test (Apti) :
1. Probability of 0 → 1 is p1 and 1 → 0 is p2. If 00 is xferred what is the prob of receiving at least one of them is 0. Ans 1 – (p1*p1)
2. Triangle 1: width is 5, height is 2.
Triangle 2: width is 8, height is 3
Rectangle 3: width is 5, height is 3
What's the total area?
a) 32
b) 32.5
c) 33
d) both a and b
2. Something like this:
6471p + 3245q = 263452
3245p + 6471q = 236231
a) 1.5 <= p <= 2
b) 2 <= p <= 2.5 etc.
3. Speed downstream is 72 Kmph, level is 63 Kmph, upstream is 54 Kmph. A person travels A to B in 4 hrs and returns in 4 hrs 40 min. Distance from A to B is
a) 203 Km
b) 273 Km
c) 302 Km
d) Data insufficient
4. Something like this:
A said "B didn't do it"
B said "I didn't do it"
C said "A did it"
D said "B lies"
Who is true?
5. Speed uphill=53miles/hr; speed downhill=70 miles/hr; speed on flat road=63miles/sec. It takes 4 hr to travel from town A to B and 4hrs 40min to travel from B to A. Find the distance between the towns.
3
1
2
6. A has n+1 coins and B has n coins. Both of them together toss all their coins. What is the probability that A gets more no of heads than B.
Interview
1. How to construct 4x1 mux using 2x1 mux only.
2. How to find out contents of PC at any point in the code. Ans Using CALL and reading top of stack.
3. x = (x +1) % 2 in the body of big loop. Optimize this to single operation. Initially x = 0. Ans x = not (x)
4. How to make a monostable (one shot) multivibrator using flipflops.
5. If the clock and D input of a D flipflop are shoted and clock connected to this circuit, how will it respond?

6. Some opamp circuit with several voltage and current sources connected through resistor dividers, find output.
7. Basic DSP theory: What is the frequency domain representation of (1) sinewave (2) cosine wave (3) the combination of sine and cosine waves. Given the output of (3)above, how will you find the input? Draw and show how it looks like.
8. If a LPF and HPF are connected in series, how will they respond under different cases of their cutoff frequencies (example if f1 < f2, what will happen)?
9. Interface an 8 bit µP with two 8Kx8 RAM chips. What would you do if A0, A1 are interchanged in h/w for only one memory chip. What'd you do in case of PROMs in case of RAMs?
DSP Paper
There are 3 sections ee, dsp and cse each with 20 q?s u have to attempt any one section only . Here I am sending the dsp section which I took , in other section the first and last 4 q?s were same as dsp.
1

--------S ----R1-------
| R2
V |___
| R3 |(C
| | -----
---
instantaneous Voltage across R2 when switch S is closed :
a.V*R2/(R1+R2)
b.V*R2/(R1+R2+R3)
c.0
d.V
Ans V*R2/(R1+R2+R3)------is what I wrote
2

---------R----------
| |
V L
| |
_____
as freq increases which of the following increases
ans : Z and V(L)
3 q? on setup time and dealy
diagram below given not very clear\
clock period is 10ns ,
setuptime of each ff is 2 ns
clk2Q delay is 3ns
Slew dew to inverter is 1ns
Wht is the max allowable dealy of block D hold time=0
Ans 10-2-3-1=4ns
4 o/p of the following gate
D
CLK=10ns Slew=1
ans : (a+b)c+de
5 SER=10^-4 the BER of a QPSK
a =SER
b <=SER
c>=SER

d =SER/2
ans >=SER
6 for 62db of PCM System what is the no of bits =10
7 for a 4 level pipeline processor the no of machine cycles required for executing 4 and (someno I don't rember) with initially pipeline flushed
ans = 4+3 and …+3
u add there for initial latency
8 An ideal LPF is
a causal
b non causal
c non stable
d none
ans: non causal ,
9 impluse func and white noise have same
a *magnitude and phase response*
b magnitude response
c phase response
d none
ans magnitude and phase response
or
nand
nand
and not
A
B
C
D
eEzxdcfgvbhnjmk,fvghbjnkl